NE523 Homework 2

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Question 1 Fluxes & Currents

Angular flux $\psi(\hat{\Omega})$ is the intensity of neutrons in the direction $\hat{\Omega}$. In this case, since the intensity is I when in direction $\hat{i}$ (the unit vector in the x-direction), and zero otherwise. So,

$$\psi(\hat{\Omega}) = I\delta(\hat{i} \cdot \hat{\Omega} - 1), \tag{1}$$

since $\hat{i} \cdot \hat{\Omega} = 1$ when $\hat{\Omega} = \hat{i}$. Then, using this angular flux,

$$\varphi = \int\limits_{4\pi} d\Omega \; I\delta(\hat{i} \cdot \hat{\Omega} - 1) = I. \tag{2}$$

The current vector is

$$\boldsymbol{J} = \int\limits_{4\pi} d\Omega \; \hat{\Omega}\psi = \hat{i}I, \tag{3}$$

and the partial currents are

$$J_x^+ = \int\limits_{\hat{\Omega}\cdot\hat{i}>0} d\Omega \; \hat{i} \cdot \hat{\Omega}\psi = I \tag{4}$$

$$J_x^- = \int\limits_{\hat{\Omega}\cdot\hat{i}<0} d\Omega \; \hat{i} \cdot \hat{\Omega}\psi = 0 \tag{5}$$

Question 2 Slab Geometry Transport Solution

The transport equation for a purely absorbing medium with isotropic source is

$$\hat{\Omega} \cdot \nabla\psi(\boldsymbol{r}, \hat{\Omega}, E) + \sigma_t\psi = \frac{1}{4\pi}q(E). \tag{6}$$

Then, since in slab geometry $\frac{d\psi}{dy} = \frac{d\psi}{dz} = 0$, this becomes

$$\mu \frac{d}{dx}\psi(x, \hat{\Omega}, E) + \sigma_t \psi = \frac{1}{4\pi} q(E). \tag{7}$$

Since this is a seperable ODE, the solution can be found by rearranging and integrating both sides:

$$\begin{aligned}
-\mu \frac{d\psi}{dx} &= \sigma_t \psi(x, \hat{\Omega}, E) - \frac{1}{4\pi} q(E) \qquad (8)\\
\frac{1}{\sigma_t \psi(x, \hat{\Omega}, E) - \frac{1}{4\pi} q(E)} d\mu &= \frac{-1}{\mu} dx \qquad (9)\\
\int \frac{1}{\sigma_t \psi(x, \hat{\Omega}, E) - \frac{1}{4\pi} q(E)} d\mu &= -\int \frac{1}{\mu} dx \qquad (10)\\
\frac{\ln\left(\sigma_t \psi(x, \hat{\Omega}, E) - \frac{1}{4\pi} q(E)\right)}{\sigma_t} &= -\frac{x}{\mu} + c_0 \qquad (11)\\
\sigma_t \psi(x, \hat{\Omega}, E) - \frac{1}{4\pi} q(E) &= c_1 e^{-\frac{\sigma_t}{\mu} x} \qquad (12)\\
\sigma_t \psi(x, \hat{\Omega}, E) &= c_1 e^{-\frac{\sigma_t}{\mu} x} + \frac{1}{4\pi} q(E) \qquad (13)\\
\psi(x, \hat{\Omega}, E) &= c_2 e^{-\frac{\sigma_t}{\mu} x} + \frac{1}{4\pi\sigma_t} q(E). \qquad (14)
\end{aligned}$$

The boundary conditions are different for different values of μ; for $\mu > 0$, the constant $c_2 = 0$ since ψ must be finite for the left half of the domain. For $\mu < 0$, the incoming flux boundary condition $\psi(0, \mu < 0) = 0$ can be used, and the constant becomes $c_2 = -\frac{1}{4\pi\sigma_t} q(E)$. Then in the slab, the flux is

$$\psi(x, E) = \begin{cases} \frac{1}{4\pi\sigma_t} q(E), & \mu > 0 \\ \frac{1}{4\pi\sigma_t} q(E)\left(1 - e^{\frac{\sigma_t}{\mu} x}\right), & \mu < 0 \end{cases}, \quad -\infty \leq x < 0. \tag{15}$$

At the interface $x = 0$, since there is no incoming flux from the vacuum,

$$\psi(0, \Omega, E) = \begin{cases} \frac{1}{4\pi\sigma_t} q(E), & \mu \geq 0 \\ 0, & \mu < 0 \end{cases} \tag{16}$$

and the current vector is

$$\boldsymbol{J}(0, E) = \int\limits_{4\pi} d\Omega\, \hat{\Omega}\psi = \hat{i}\frac{1}{2} q(E) \tag{17}$$

Question 3 Inner Products & Orthogonality

For the set of polynomials to be orthonormal on $x \in [-1, 1]$, they must satisfy the condition that $\langle P_i, P_j \rangle = \int_{-1}^{1} P_i P_j dx = \delta_{i,j}$:

$$\begin{aligned}
\langle P_0, P_0 \rangle &= \int\limits_{-1}^{1} \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} dx \\
&= \frac{1}{2} \int\limits_{-1}^{1} dx \\
\langle P_0, P_0 \rangle &= \boxed{1}
\end{aligned} \tag{18}$$

$$\begin{aligned}
\langle P_0, P_1 \rangle &= \int\limits_{-1}^{1} \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{\sqrt{2}} x dx \\
&= \frac{1}{2} \int\limits_{-1}^{1} x dx \\
&= \frac{1}{2} \left[\frac{x^2}{2} \right]_{-1}^{1} \\
&= \frac{1}{2} \left[\frac{1}{2} - \frac{1}{2} \right] \\
\langle P_0, P_1 \rangle &= \boxed{0}
\end{aligned} \tag{19}$$

$$\begin{aligned}
\langle P_0, P_2 \rangle &= \int\limits_{-1}^{1} \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{5}}{\sqrt{2}} \cdot \frac{1}{2} \left(3x^1 - 1 \right) dx \\
&= \frac{\sqrt{5}}{4} \int\limits_{-1}^{1} \left(3x^2 - 1 \right) dx \\
&= \frac{\sqrt{5}}{4} \left[x^3 - x \right]_{-1}^{1} \\
&= \frac{\sqrt{5}}{4} \left[(1 - 1) - (1 - 1) \right] \\
\langle P_0, P_2 \rangle &= \boxed{0}
\end{aligned} \tag{20}$$

$$\begin{aligned}
\langle P_1, P_1 \rangle &= \frac{3}{2}\int\limits_{-1}^{1} x^2 dx \\
&= \frac{3}{2}\left[\frac{x^3}{3}\right]_{-1}^{1} \\
&= \frac{3}{2}\left[\frac{1}{3} - \frac{-1}{3}\right]_{-1}^{1} \\
\langle P_1, P_1 \rangle &= \boxed{1}
\end{aligned} \tag{21}$$

$$\begin{aligned}
\langle P_1, P_2 \rangle &= \sqrt{\frac{3}{2}}\sqrt{\frac{5}{2}}\int\limits_{-1}^{1} \frac{1}{2}x\left(3x^2 - 1\right) dx \\
&= \frac{\sqrt{15}}{2}\int\limits_{-1}^{1} \frac{3}{4}x^2 - \frac{1}{2}x \, dx \\
&= \frac{\sqrt{15}}{2}\left[\frac{3}{8}x^4 - \frac{1}{4}x^2\right]_{-1}^{1} \\
&= \frac{\sqrt{15}}{2}\left[\left(\frac{3}{8} - \frac{1}{4}\right) - \left(\frac{3}{8} - \frac{1}{4}\right)\right] \\
\langle P_1, P_2 \rangle &= \boxed{0}
\end{aligned} \tag{22}$$

$$\begin{aligned}
\langle P_2, P_2 \rangle &= \frac{5}{2}\int\limits_{-1}^{1} \frac{1}{2^2}\left(3x^2 - 1\right)^2 dx \\
&= \frac{5}{8}\int\limits_{-1}^{1} 9x^4 - 6x^2 + 1 \, dx \\
&= \frac{5}{8}\left[\frac{9}{5}x^5 - 2x^3 + x\right]_{-1}^{1} \\
&= \frac{5}{8}\left[\left(\frac{9}{5} - 2 + 1\right) - \left(\frac{-9}{5} + 2 - 1\right)\right] \\
\langle P_2, P_2 \rangle &= \boxed{1}.
\end{aligned} \tag{23}$$

This confirms the condition that $\langle P_i, P_j \rangle = \delta_{i,j}$, so the set of polynomials is orthonormal.

Question 4 Projection on Orthogonal Basis

Any quantity (function, vector, etc) can be represented as a linear conbination of orthonormal basis componenents, with coefficients equal to the magnitude of a projection onto those bases. For functions on $x \in [-1, 1]$, using the Legendre polynomials as a basis, this expansion is

$$f(x) = \sum_{l}^{\infty} \frac{\langle f, P_l \rangle}{\langle P_l, P_l \rangle} P_l. \tag{24}$$

since $\langle P_l, P_l \rangle = 1$, the coefficients ("moments") are $f_l = \langle f, P_l \rangle \, ; l = 0, \ldots, \infty$. Assuming that the series $\sum_{l=0}^{L} f_l P_l$ converges to $f(x)$ (that is, each successive term in the sequence of truncated reconstructions is closer to the true function than the previous term), the function can be approximated using only the first $L + 1$ terms.

4(a) The Legendre moments are

$$f_i = \langle f(x), P_i(x) \rangle = \int_{-1}^{1} . \tag{25}$$

Then, for $f(x) = 3x + 7$,

$$\begin{aligned}
f_0 &= \int_{-1}^{1} \frac{1}{\sqrt{2}} (3x + 7) \, dx \\
&= \frac{1}{\sqrt{2}} \left[\frac{3}{2} x^2 + 7x \right]_{-1}^{1} \\
&= \frac{1}{\sqrt{2}} \left[\left(\frac{3}{2} + 7 \right) - \left(\frac{3}{2} - 7 \right) \right] \\
&= \frac{1}{\sqrt{2}} [14] \\
&= \frac{14}{\sqrt{2}} \\
f_0 &= 7\sqrt{2}
\end{aligned} \tag{26}$$

$$\begin{aligned}
f_1 &= \frac{\sqrt{3}}{\sqrt{2}} \int_{-1}^{1} x \, (3x + 7) \, dx \\
&= \frac{\sqrt{3}}{\sqrt{2}} \left[x^3 + \frac{7}{2} x^2 \right]_{-1}^{1} \\
&= \frac{\sqrt{3}}{\sqrt{2}} \left[\left(1 + \frac{7}{2} \right) - \left(-1 + \frac{7}{2} \right) \right] \\
&= \frac{2\sqrt{3}}{\sqrt{2}} \\
f_1 &= \sqrt{6}
\end{aligned} \tag{27}$$

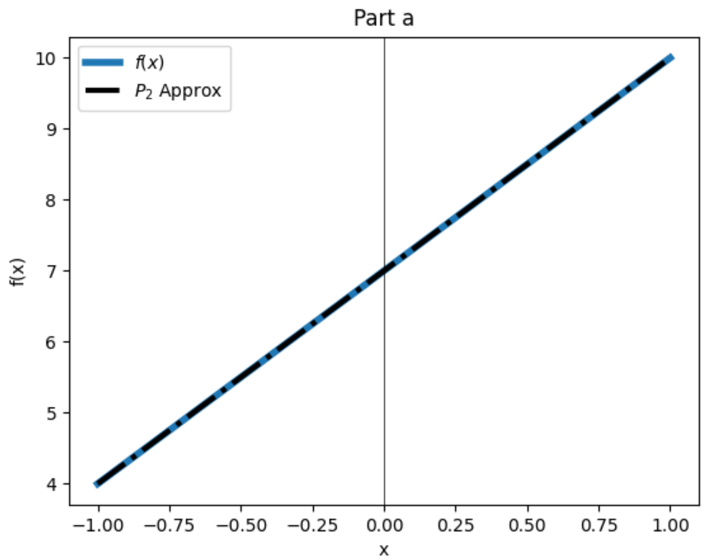

Figure 1: Function and approximation, Part a

$$\begin{aligned} f_2 &= \frac{\sqrt{5}}{\sqrt{2}} \int_{-1}^{1} \frac{1}{2}\left(3x^2 - 1\right)(3x+7)\,dx \\ &= \frac{1}{2}\frac{\sqrt{5}}{\sqrt{2}} \int_{-1}^{1} \left(9x^3 - 3x + 21x^2 - 7\right)dx \\ &= \frac{1}{2}\frac{\sqrt{5}}{\sqrt{2}} \left[\frac{9}{4}x^2 - \frac{3}{2}x^2 + 7x^3 - 7x\right]_{-1}^{1} \\ f_2 &= 0 \end{aligned} \tag{28}$$

Because the original function is a polynomial with degree less than two, the truncated Legendre expansion is equal to the original funciton, as seen in Figure 1.

4(b) For the function $f(x) = e^x$,

$$\begin{aligned} f_0 &= \frac{1}{\sqrt{2}} \int_{-1}^{1} e^x dx \\ &= \frac{1}{\sqrt{2}} \left[e^x\right]_{-1}^{1} \\ f_0 &= \frac{1}{\sqrt{2}} \left[e^1 - e^{-1}\right] \end{aligned} \tag{29}$$

$$\begin{aligned}
f_1 &= \sqrt{\frac{3}{2}} \int\limits_{-1}^{1} x e^x dx \\
&= \sqrt{\frac{3}{2}} \left[x e^x \big|_{-1}^{1} - \int\limits_{-1}^{1} e^x dx \right] \\
&= \sqrt{\frac{3}{2}} \left[\left(e^1 + e^{-1} \right) - \left(e^1 - e^{-1} \right) \right] \\
f_1 &= \sqrt{6} e^{-1}
\end{aligned} \tag{30}$$

$$\begin{aligned}
f_2 &= \frac{1}{2}\sqrt{\frac{5}{2}} \int\limits_{-1}^{1} \left(3x^2 - 1\right) e^x dx \\
&= \frac{1}{2}\sqrt{\frac{5}{2}} \left[e^x \left(3x^2 - 1\right) \big|_{-1}^{1} - \int\limits_{-1}^{1} 6x e^x dx \right] \\
&= \frac{1}{2}\sqrt{\frac{5}{2}} \left[\left[e^x \left(3x^2 - 1\right) - 6x e^x \right]_{-1}^{1} + \int\limits_{-1}^{1} 6 e^x dx \right] \\
&= \frac{1}{2}\sqrt{\frac{5}{2}} \left[e^x \left(3x^2 - 6x - 1 + 6\right) \right]_{-1}^{1} \\
&= \frac{1}{2}\sqrt{\frac{5}{2}} \left[e^x \left(3x^2 - 6x + 5\right) \right]_{-1}^{1} \\
&= \frac{1}{2}\sqrt{\frac{5}{2}} \left[e^1 (3 - 6 + 5) - e^{-1} (3 + 6 + 5) \right] \\
f_2 &= \frac{1}{2}\sqrt{\frac{5}{2}} \left[2e^1 - 14e^{-1} \right]
\end{aligned} \tag{31}$$

Though the function is exponential, the truncated Legendre approximation is still close with only three moments used, as seen in Figure 2.

4(c) For the function $f(x) = e^{5x}$,

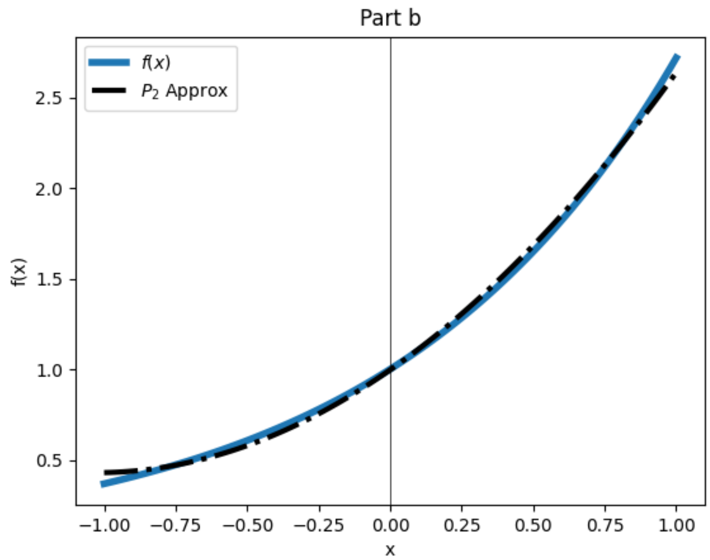

Figure 2: Function and approximation, Part b

$$
\begin{aligned}
f_0 &= \frac{1}{\sqrt{2}} \int_{-1}^{1} e^{5x} dx \\
&= \frac{1}{\sqrt{2}} \left[\frac{1}{5} e^{5x} \right]_{-1}^{1} \\
f_0 &= \frac{1}{5\sqrt{2}} \left[e^{5} - e^{-5} \right]
\end{aligned}
\tag{32}
$$

$$
\begin{aligned}
f_1 &= \sqrt{\frac{3}{2}} \int_{-1}^{1} x e^{5x} dx \\
&= \sqrt{\frac{3}{2}} \left[\frac{1}{5} x e^{5x} \bigg|_{-1}^{1} - \int_{-1}^{1} \frac{1}{5} e^{5x} dx \right] \\
&= \sqrt{\frac{3}{2}} \left[\frac{1}{5} x e^{5x} - \frac{1}{25} e^{5x} \right]_{-1}^{1} \\
&= \frac{1}{5} \sqrt{\frac{3}{2}} \left[\left(e^{5} - \frac{1}{5} e^{-5} \right) - \left(-e^{5} - \frac{1}{5} e^{-5} \right) \right] \\
f_1 &= \frac{1}{25} \sqrt{\frac{3}{2}} \left[4e^{5} + 6e^{-5} \right]
\end{aligned}
\tag{33}
$$

$$
\begin{aligned}
f_2 &= \frac{1}{2}\sqrt{\frac{5}{2}}\int_{-1}^{1} e^{5x}\left(3x^2-1\right)dx \\
&= \frac{1}{2}\sqrt{\frac{5}{2}}\left[\int_{-1}^{1} 3x^2e^{5x}dx - \int_{-1}^{1} e^{5x}dx\right] \\
&= \frac{1}{2}\sqrt{\frac{5}{2}}\left[\frac{3}{5}x^2e^{5x}\Big|_{-1}^{1} - \int_{-1}^{1}\frac{6}{5}xe^{5x}dx - \frac{1}{5}e^{5x}\Big|_{-1}^{1}\right] \\
&= \frac{1}{2}\sqrt{\frac{5}{2}}\left[\left(\frac{3}{5}x^2e^{5x} - \frac{6}{25}xe^{5x} - \frac{1}{5}e^{5x}\right)_{-1}^{1} + \int_{-1}^{1}\frac{6}{25}e^{5x}dx\right] \\
&= \frac{1}{2}\sqrt{\frac{5}{2}}\left[\frac{3}{5}x^2e^{5x} - \frac{6}{25}xe^{5x} - \frac{1}{5}e^{5x} + \frac{6}{125}e^{5x}\right]_{-1}^{1} \\
&= \frac{1}{2}\sqrt{\frac{5}{2}}\left[\left(\frac{75}{125}x^2 - \frac{30}{125}x + \frac{19}{125}\right)e^{5x}\right]_{-1}^{1} \\
&= \frac{1}{250}\sqrt{\frac{5}{2}}\left[\left(75x^2-30x+19\right)e^{5x}\right]_{-1}^{1} \\
&= \frac{1}{250}\sqrt{\frac{5}{2}}\left[(75-30-19)\,e^{5} - (75+30-19)\,e^{-5}\right] \\
f_2 &= \frac{1}{250}\sqrt{\frac{5}{2}}\left[26e^{5} - 86e^{-5}\right]
\end{aligned}
\tag{34}
$$

Because of the rapid growth of e^{5x}, this function is poorly approximated with polynomials, as seen in Figure 3.

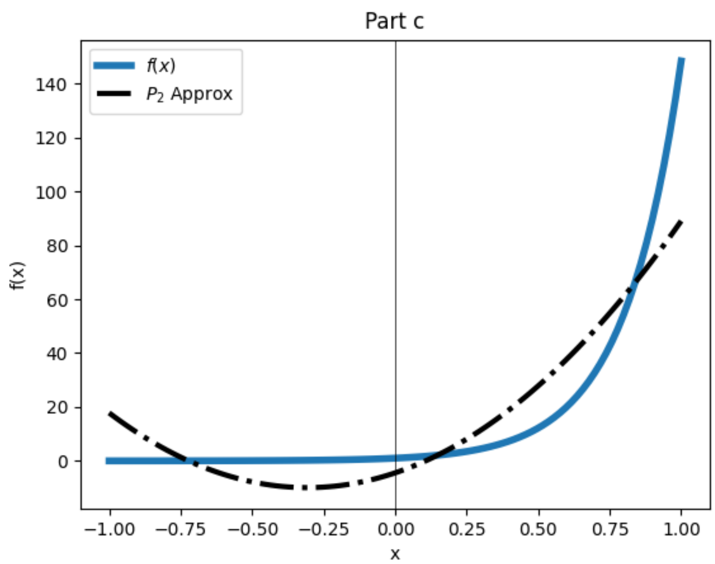

Figure 3: Function and approximation, Part c